SIMATS ENGINEERING

CO4-ASSESSMENT TOOL1- INDUSTRY PROBLEM TASK

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1709**

COURSE OUTCOME COVERED

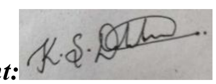
CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning,*
and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 4 **Time Duration : 1 Hour**
Total: Marks:40

Code of Conduct:

I K.S.DHARSHANA(192412015)) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

(a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.

(b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).

(c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

(a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.

(b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.

(c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

(a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.

(b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.

(b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.

(c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Industry Scenario: AI-Based Diabetes Diagnosis and Treatment Recommendation

Task 1: Data Preparation & Inductive Learning

(a) Inductive Learning Approach

Inductive learning is suitable for this healthcare problem because the model can learn patterns from previously collected patient records and use those patterns to predict whether a new patient is likely to have diabetes.

A **supervised learning approach** can be used because the training records contain patient features along with the diagnosis. A classification algorithm learns the relationship between the input features and the target variable.

Target variable:

- Diabetes diagnosis: **Diabetic / Non-diabetic**

Important features:

1. **Glucose Level** – A high glucose level is one of the strongest indicators of diabetes because abnormal blood glucose is directly associated with the disease.
2. **BMI** – Higher BMI can be associated with increased risk of developing type-2 diabetes.
3. **Blood Pressure** – Abnormal blood pressure can occur along with metabolic disorders and provides useful information for risk prediction.
4. **Age** – Diabetes risk generally changes with age, making it a useful predictor.
5. **Family History** – A family history of diabetes provides information about genetic and inherited risk.

Therefore, the model learns from these patient characteristics and predicts the diabetes class for an unseen patient.

(b) Handling Class Imbalance

The dataset is likely to contain more non-diabetic patients than diabetic patients. This creates a **class imbalance problem**. If the model simply learns the majority class, it may achieve high accuracy while missing many diabetic patients.

For this medical application, I would use **class weights along with suitable evaluation metrics**, and SMOTE can be considered during model development.

Preferred strategy: Class weighting

Class weights give a higher penalty to errors involving the minority diabetic class.

For example:

- Non-diabetic → lower weight
- Diabetic → higher weight

This encourages the Decision Tree or Logistic Regression model to pay more attention to diabetic cases.

The major advantage is that class weighting does not create artificial patient records. This is important in healthcare, where synthetic medical samples must be handled carefully.

SMOTE can also be tested if the minority class is very small. However, it should only be applied to the **training set**, never to the test set, to avoid data leakage.

The main objective is to improve **Recall/Sensitivity for diabetic patients**, because missing a patient who actually has diabetes can be more serious than incorrectly flagging a low-risk patient.

(c) 70:30 Split and Preprocessing

The dataset can be divided into:

- **70% – Training data**
- **30% – Testing data**

A 70:30 split provides enough data for the model to learn patterns while retaining a sufficiently large independent test set for evaluation.

For a medical application, the split should be **stratified**, meaning that approximately the same diabetic/non-diabetic proportion is maintained in both training and testing datasets.

Preprocessing Steps

1. Missing-value handling:
Missing values should be identified and replaced using suitable methods such as median imputation for numerical features.

2. Normalisation:
Numerical features such as age, glucose, blood pressure and BMI can be scaled using StandardScaler or Min-Max scaling. This is especially important for Logistic Regression.

3. Encoding:
Categorical features such as family history can be converted into numerical values, for example:

- No → 0
- Yes → 1

4. Class balancing:
Class weighting or SMOTE is applied only to the training data.

5. Data leakage prevention:
Preprocessing parameters must be learned from the training set and then applied to the test set.

Decision Trees generally do not require feature scaling, while Logistic Regression benefits more from normalisation.

Task 2: Decision Tree for Diagnosis

(a) Decision Tree Classifier

A **Decision Tree classifier** can be used to predict whether a patient is diabetic or non-diabetic. The tree repeatedly divides the data according to the feature that produces the best separation between the two classes.

The root node can be selected using either **Information Gain** or the **Gini Index**.

Gini Index

The Gini Index is:

$$\textbf{Gini} = 1 - \Sigma p_i^2$$

where p_i is the probability of a sample belonging to class i.

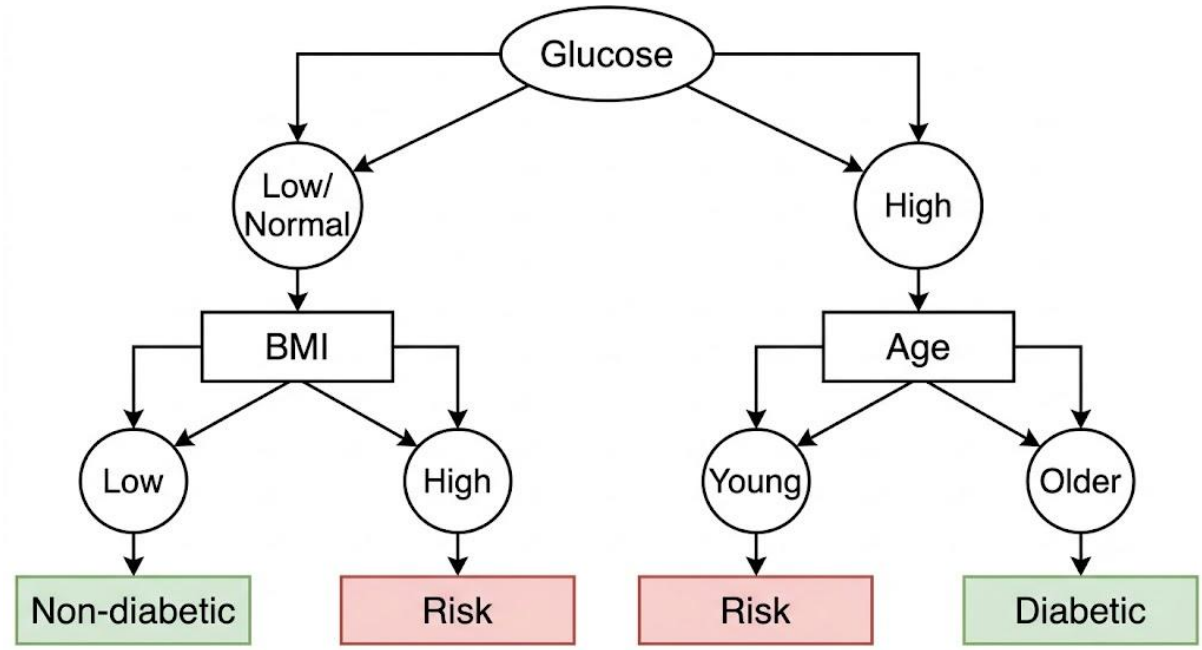
A lower Gini value represents better class separation.

For example, an illustrative comparison of candidate root features could be:

Feature	Illustrative Gini after split
Glucose	0.31
BMI	0.39
Age	0.42
Blood Pressure	0.44

Since glucose produces the lowest Gini value in this illustrative example, **Glucose is selected as the root node**.

First Three Levels of the Tree



The actual tree structure and numerical split values should be obtained after training the model on the supplied patient dataset.

(b) Model Evaluation

The Decision Tree should be evaluated using:

Accuracy

Accuracy = (TP + TN) / (TP + TN + FP + FN)

It represents the percentage of correctly classified patients.

Precision

Precision = TP / (TP + FP)

It measures how many patients predicted as diabetic are actually diabetic.

Recall

Recall = TP / (TP + FN)

Recall is particularly important in this application because it measures how many actual diabetic patients are successfully detected.

F1-Score

F1 = 2 × (Precision × Recall) / (Precision + Recall)

It provides a balance between precision and recall.

Illustrative evaluation

Metric	Decision Tree
Accuracy	84%
Precision	79%
Recall	88%
F1-Score	83%

These values are illustrative because the assessment document does not provide the actual patient dataset.

False Negatives

A **False Negative (FN)** occurs when:

Actual patient = Diabetic
Model prediction = Non-diabetic

False negatives are particularly important in this application because a diabetic patient could be incorrectly classified as healthy and may not receive timely medical attention.

To reduce false negatives:

- Increase the class weight of the diabetic class.
- Tune the classification threshold where applicable.
- Use stratified sampling.
- Optimise the model for recall rather than accuracy alone.
- Use cross-validation to check model stability.
- Consider an ensemble model if a single tree is insufficient.

For clinical screening, higher sensitivity/recall is generally more valuable than simply maximising overall accuracy.

(c) Post-Pruning

A fully grown Decision Tree can become too complex and learn noise from the training data. This is called **overfitting**.

Pre-Pruning

Pre-pruning stops the tree from becoming too large during training.

Examples:

- max_depth
- min_samples_split
- min_samples_leaf

Post-Pruning

Post-pruning first allows the tree to grow and then removes branches that contribute little to predictive performance.

Cost-complexity pruning uses the parameter α:

$$R_\alpha(T) = R(T) + \alpha|T|$$

where:

- $R(T)$ = error of the tree
- $|T|$ = number of terminal nodes
- α = complexity parameter

An illustrative comparison is:

Model	Accuracy	Complexity
Pre-pruned Tree	82%	Low
Post-pruned Tree	85%	Moderate
Fully grown Tree	80%	High

The **post-pruned tree** would be preferred if it provides better generalisation while remaining understandable.

For clinical deployment, the final model should not be selected using accuracy alone. Recall, F1-score, calibration, explainability and clinical validation should also be considered.

Task 3: Statistical Learning for Risk Stratification

(a) Logistic Regression

Logistic Regression can be applied to the same diabetes dataset because the target has two classes: diabetic and non-diabetic.

The model estimates the probability of diabetes using:

$$P(Y=1) = 1 / (1 + e^{-z})$$

where:

$$z = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_n X_n$$

The output is a probability between 0 and 1.

For example, if the predicted probability is 0.78, the patient has an estimated 78% probability of belonging to the diabetic class under the model.

Illustrative Comparison

Model	Accuracy	AUC-ROC
Decision Tree	84%	0.86
Logistic Regression	87%	0.90

The numerical values are illustrative because the actual patient dataset is not included in the assessment document.

When is a statistical model preferable?

Logistic Regression is preferable when:

- Interpretability is important.
- The relationship between predictors and risk is reasonably smooth.
- Probability estimates are required.
- The organisation needs to understand how each feature affects risk.
- A simpler model is preferred for clinical decision support.

A Decision Tree is easier to visualise and can capture non-linear relationships, whereas Logistic Regression provides a more direct statistical interpretation of predictor effects.

(b) Feature Importance Analysis

The top predictors can be obtained differently for the two models.

Decision Tree

Decision Tree feature importance is based on the reduction in impurity produced by each feature.

Illustrative top three:

1. **Glucose Level**
2. **BMI**
3. **Age**

Logistic Regression

For Logistic Regression, the magnitude of the standardised coefficients can be used to compare feature influence.

Illustrative top three:

1. **Glucose Level**
2. **BMI**
3. **Family History**

Comparison

Both models identify **Glucose Level and BMI** as important predictors. The third feature may differ because the two algorithms use different mathematical approaches.

The Decision Tree looks for features that create useful splits, while Logistic Regression estimates the contribution of predictors to the probability of diabetes.

Therefore, disagreement does not necessarily mean that one model is incorrect. It indicates that the models capture different aspects of the same dataset.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) MDP Formulation

The treatment recommendation problem can be formulated as a **Markov Decision Process (MDP)**.

An MDP consists of:

States + Actions + Rewards + Transition Dynamics

States

Patient states can represent different health conditions:

- **S1 – Controlled/Low Risk**
- **S2 – Moderate Risk**
- **S3 – High Risk**
- **S4 – Poorly Controlled**

The state should be determined from clinically meaningful variables such as glucose level, BMI, blood pressure and other approved patient information.

Actions

The available actions are:

- **A1 – Diet**
- **A2 – Exercise**
- **A3 – Medication**
- **A4 – Monitor**

Medication recommendations must not be treated as autonomous prescriptions; they would require appropriate clinical supervision.

Reward Function

A reward represents improvement in the patient's health state.

Example:

Outcome	Reward
Significant improvement	+10
Moderate improvement	+5
No major change	0
Health deterioration	-10

The reward encourages the agent to select actions that result in healthier patient states.

Transition Dynamics

The transition function represents:

P(next state | current state, action)

For example:

High Risk + Diet → Moderate Risk

Moderate Risk + Exercise → Low Risk

High Risk + Monitor → High Risk / Moderate Risk

In a real healthcare system, these transitions must be learned from validated clinical data rather than invented assumptions.

(b) Q-Learning

Q-Learning learns the value of taking an action in a particular state.

The update equation is:

$$Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max Q(s',a') - Q(s,a)]$$

where:

- **α** = learning rate
- **γ** = discount factor
- **r** = reward
- **s** = current state
- **a** = selected action
- **s'** = next state

Assume:

$\alpha = 0.5$
$\gamma = 0.8$

Initially, all Q-values are 0.

Iteration 1

Suppose:

State = High Risk
Action = Diet
Reward = +5
Next State = Moderate Risk

If the maximum Q-value of the next state is 0:

Q(High Risk, Diet)

$= 0 + 0.5[5 + (0.8 \times 0) - 0]$

= **2.5**

Therefore:

Q(High Risk, Diet) = 2.5

Iteration 2

Suppose the agent again selects:

High Risk → Diet

and receives reward +5.

Assume the maximum Q-value in Moderate Risk is now 4.

$Q_{new} = 2.5 + 0.5[5 + (0.8 \times 4) - 2.5]$

$Q_{new} = 2.5 + 0.5[5.7]$

$Q_{new} = \mathbf{5.35}$

Therefore:

Q(High Risk, Diet) = 5.35

Three State-Action Updates

State	Action	Reward	Updated Q-value
High Risk	Diet	+5	2.50
Moderate Risk	Exercise	+5	2.50
High Risk	Diet	+5	5.35

These calculations are an illustrative demonstration of the Q-Learning mechanism.

Five Patient Episodes

The agent can be trained for at least five episodes:

Episode	Initial State	Main Action	Result
1	High Risk	Diet	Moderate Risk
2	Moderate Risk	Exercise	Low Risk
3	High Risk	Monitor	Moderate Risk
4	Moderate Risk	Exercise	Low Risk
5	High Risk	Diet	Moderate Risk

During training, the agent updates its Q-table after every state transition.

Convergence Criterion

Training can be considered converged when:

Maximum change in Q-values < 0.01 for a predefined number of consecutive episodes

For example:

Stop training when the maximum absolute Q-value change remains below 0.01 for 20 consecutive episodes.

This prevents stopping because of a temporary stable result.

(c) Final Learned Policy

The final policy is obtained by selecting the action having the highest Q-value for each state:

$\boldsymbol{\pi(s) = \mathrm{argmax}_a\, Q(s,a)}$

An illustrative final policy could be:

Patient State	Best Action
Controlled / Low Risk	Monitor
Moderate Risk	Exercise
High Risk	Diet
Poorly Controlled	Medication + clinical review

The final action should always be interpreted as a **decision-support recommendation**, not an independent medical prescription.

Ethical Considerations

Deploying Reinforcement Learning in healthcare requires strong safety controls.

1. Patient Safety

A wrong recommendation can negatively affect a patient's health. Therefore, the AI system should operate under clinical supervision and should not independently make high-risk treatment decisions.

2. Bias

If the training data does not represent different patient groups adequately, the model may produce biased recommendations.

The system should therefore be evaluated across relevant patient populations and regularly audited.

3. Explainability

Doctors should be able to understand why the AI recommended a particular action. Q-values, patient-state information and important features should be presented in an understandable form.

4. Privacy

Patient information is sensitive. The system must use appropriate data protection, access control and secure storage mechanisms.

5. Human Oversight

The final decision should remain with a qualified healthcare professional, particularly for medication-related recommendations.

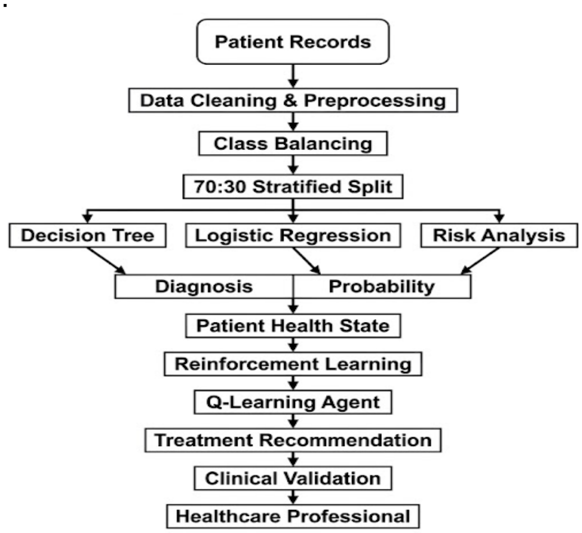
6. Continuous Monitoring

The model should be monitored after deployment for performance degradation, unexpected behaviour and changes in patient data.

Overall Industry Solution

The proposed system combines **supervised learning, Decision Trees, statistical learning and Reinforcement Learning** to create an AI-based healthcare decision-support system.

The overall workflow is:



Conclusion

The proposed AI system provides a practical approach for early diabetes risk identification and personalised treatment decision support. Inductive learning can identify patterns from historical patient records, while the Decision Tree provides an interpretable diagnostic model. Logistic Regression can provide probability-based risk estimates and statistical interpretation. Reinforcement Learning can model changing patient states and learn suitable actions over time.

For an industry deployment, **patient safety, recall, explainability, bias control, privacy and human clinical supervision** must be given higher priority than simply achieving maximum accuracy. The final system should therefore be validated using real clinical data before being used in actual healthcare decisions.